

- Goal: Train an LLM to generate deeper, more nuanced storytelling.
- Constraints:
 - 5 numerical values in the framework (representing aspects of storytelling quality): Emotion provocation, empathy, engagement, authenticity, narrative complexity
 - Computational cost considerations.
 - Need for interpretability & justification in method selection.
- Possible Approaches:
 - 1. Supervised Fine-Tuning (SFT):**
 - Works well if we have **high-quality** story samples with clear, human-labeled attributes. **What's the definition of high-quality?**
 - Example-based training: manually curate a dataset of **good vs. bad** storytelling.
 - Pros: Less hyperparameter sensitivity, leverages pretrained models effectively, works well with high-quality data, stable and deterministic
 - Cons: Expensive and requires a well-labeled dataset, does not directly use preference optimization for human preferences, learns from absolute examples but does not compare different generations,
 - Conclusion:
 - Since SFT can make it less effective if story depth is a more subjective or multi-dimensional concept, we decided not to use this approach.
 - If deeper storytelling involves multiple nuanced attributes (such as engagement), then a single-label SFT approach might struggle and multi-attribute reward modeling could be better suited.
 - Training a model with a large supervised dataset can be costly in terms of GPU time and storage and preference optimization methods often require fewer training steps.
 - **Modified dataset for SFT uploaded in Github.**
 - Math:
 - SFT is a MLE-based approach that optimizes for likelihood over high-quality stories:

$$\mathcal{L}_{\text{SFT}}(\theta) = - \sum_{i=1}^N \log p_{\theta}(y_i | x_i)$$

Chain rule:

$$p_{\theta}(y|x) = \prod_{t=1}^T p_{\theta}(y_t | x, y_{<t})$$

where y_t is the token at time step t , and $y_{<t}$ represents all previous tokens.

$$\mathcal{L}_{\text{SFT}}(\theta) = - \sum_{i=1}^N \sum_{t=1}^{T_i} \log p_{\theta}(y_{i,t} | x_i, y_{i,<t})$$

Gradient Update:

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}_{\text{SFT}}(\theta)$$

Note: Since the goal is to optimize for *deeper storytelling*, we could weight examples by a quality score s_i (e.g., an expert-assigned score or an automated metric like PDS):

$$\mathcal{L}_{\text{weighted SFT}}(\theta) = - \sum_{i=1}^N s_i \sum_{t=1}^{T_i} \log p_{\theta}(y_{i,t} | x_i, y_{i,<t})$$

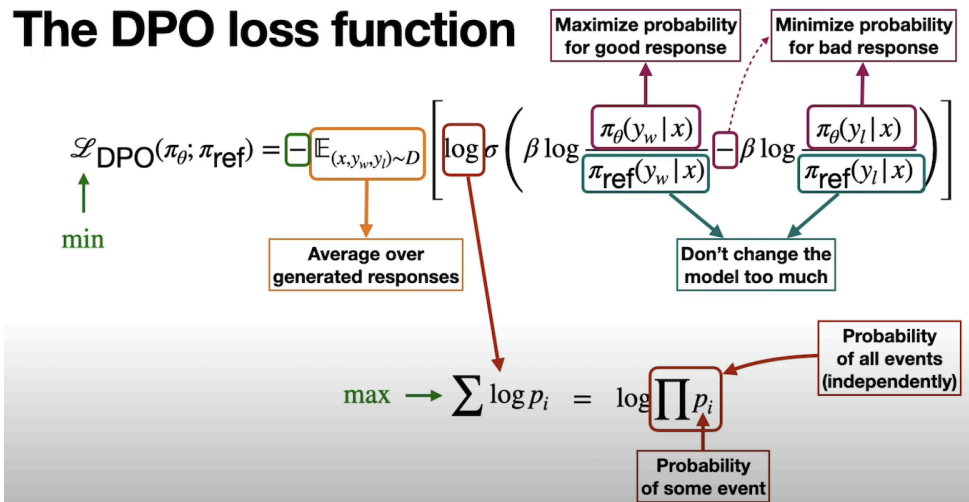
2. Direct Preference Optimization (DPO):

- DPO is a ranking-based fine-tuning method that learns directly from preference pairs, rather than absolute likelihoods (as in SFT). Instead of just maximizing the likelihood of a single "good" response, it optimizes the model to prefer better generations over worse ones.
- Pros:
 - i. Emotion, empathy, and engagement are hard to quantify explicitly, but humans can often compare two stories and say which one is better.
 - ii. Instead of needing absolute scores, DPO only requires preference rankings (e.g., "Story A provoked stronger emotions than Story B"). Since DPO is constrained to stay **close to the base model**, it avoids extreme shifts.
- Cons:
 - i. DPO assumes clear differences between "good" and "bad" outputs, but storytelling depth is often a gradient rather than binary and also it is highly subjective and can vary between different annotators. If two stories are similarly engaging but in different ways, DPO might struggle to rank them correctly.
 - ii. This is problematic for subtle aspects like authenticity—where differences may not be as stark.

- iii. Since DPO forces the model to align with past preferences, it might get stuck in a specific storytelling style rather than exploring diverse ways of being engaging.
Example: If early ranking data favors emotional depth, DPO might over-prioritize emotional storytelling at the cost of complexity or authenticity.
- o Conclusion: Since DPO is good in the scenarios where:
 - i. we want a method that does not require training a separate reward model or
 - ii. we can ensure stronger negative examples to make ranking meaningful
 - iii. We can easily generate or collect preference rankings.
 - iv. We want a data-efficient method that doesn't require absolute labels.
 - v. We are fine with relative ranking rather than explicit scores.

We assume that DPO can be a good baseline to start with. For better results, we can use a **Hybrid Approach** - use SFT first on high-quality examples, then use DPO to refine rankings.

- Math:



β is a temperature parameter that controls how sharp the preference signal is.

3. Reinforcement Learning from Human Feedback (RLHF):

- o could help in deep storytelling by optimizing emotion provocation, empathy, engagement, authenticity, and narrative complexity in a more nuanced and fine-grained way than DPO or SFT alone.
- o Unlike SFT, where the model just mimics high-quality outputs, RLHF actively improves generation based on human judgment.

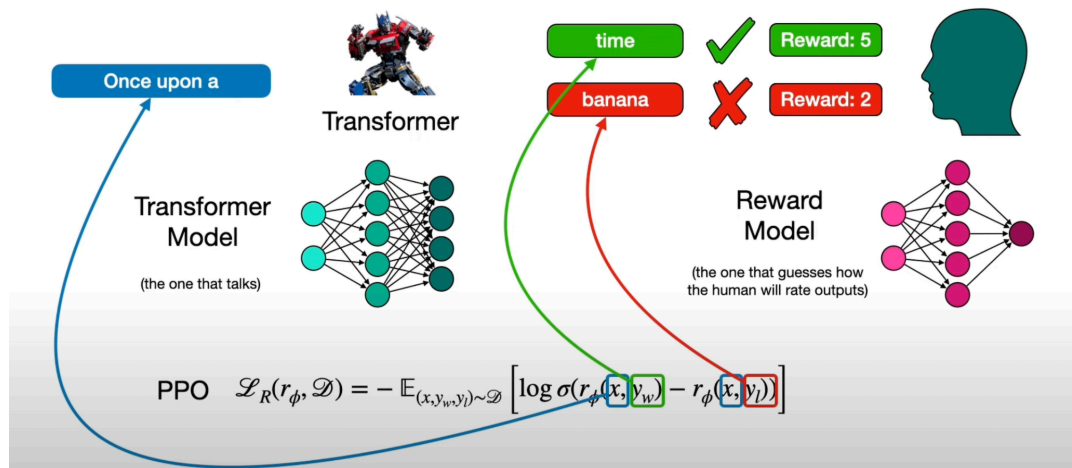
- SFT only learns from existing high-quality stories, meaning it cannot improve beyond the dataset.
- RLHF can explore novel storytelling styles and generate new, deeper stories that were not explicitly present in training data.
- If some attributes conflict (e.g., authenticity vs. engagement), RLHF can find an optimal balance by learning from human preference weightings.
- Unlike DPO, which forces the model to follow past rankings, RLHF allows for dynamic adaptation based on updated feedback.
- Pros:
 - i. RLHF allows multi-attribute reward modeling, meaning it can balance aspects like engagement vs. authenticity vs. complexity rather than optimizing just one at a time.
 - ii. Unlike DPO, which works best with clear preference pairs, RLHF can learn more continuous reward signals, helping with subtle differences in storytelling quality.
 - iii. Actively learns and improves beyond static examples.
- Cons:
 - i. Computationally expensive
 - ii. The reward model is trained on human annotations, which can introduce biases (e.g., if annotators favor simplistic but emotional stories, RLHF might overfit to that). Also Scoring deep storytelling is extremely subjective, making it hard to create consistent reward signals.
 - iii. If human annotators disagree on what makes a story engaging, the reward model can become noisy, leading to unstable training.
- Conclusion:

We decided not to use RLHF since it is computationally heavy for a baseline and also we can first try lighter approaches such as DPO and SFT and then try using RLHF when necessary. In that case, we use RLHF this way:

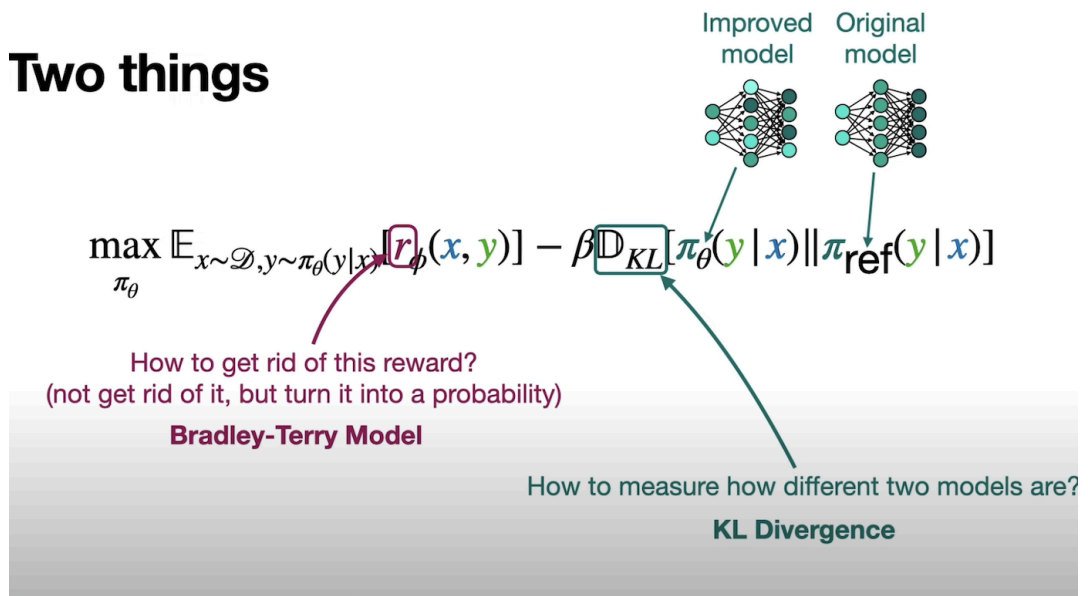
- Start with SFT → Train on high-quality deep stories to establish a good foundation.
- Use DPO → Rank pairs of stories to refine storytelling depth with preference-based learning.
- Apply RLHF → Train a reward model based on multi-attribute feedback and fine-tune for deeper storytelling.

■ Math:

Reinforcement Learning with Human Feedback (RLHF)



Two things



4. KL-preference-based Training Objective (KTO):

- Unlike DPO, which assumes strong preferences, KTO allows for soft, nuanced rankings, making it better for subjective attributes like authenticity and empathy in storytelling.
- Some stories may be close in quality but different in style (e.g., a highly engaging but less authentic story vs. a deeply authentic but slow-paced one). KTO allows partial ranking signals, whereas DPO forces a strict binary ranking.
- Unlike RLHF, which requires an explicit reward model, KTO just needs pairs of stories ranked based on their quality (which can be sourced from existing datasets or weak model generations).
- Pros:

- Because KTO smooths out the preference optimization process, it reduces **reward hacking**, where the model exploits weaknesses in the ranking system.
 - This ensures the model **doesn't learn to generate overly formulaic or exaggerated emotional responses** just to maximize scores.
- Cons:
 - Still needs good ranking data, might limit creativity
 - KTO keeps the fine-tuned model **close to the base model distribution**, which is good for **stability** but might prevent the model from learning **radically new storytelling techniques**.
- How to use KTO?
 - Use a **strong model (like GPT-4, Mistral, or LLaMA-3)** to generate multiple versions of a story with different depths.
 - Use a **weaker model** (like LLaMA-2-7B) to generate additional baseline outputs.
 - Rank these stories based on **emotion provocation, empathy, engagement, authenticity, and complexity** (either **human-labeled** or via **automatic scoring**).
- Conclusion: It might be a good idea to try using KTO as a baseline.